Baryon Acoustic Oscillations as a Frozen Coherence Imprint

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Abstract

Baryon acoustic oscillations are conventionally described as the fossil remnant of pressure-supported transport in the coupled baryon-photon plasma of the early universe, later retained as a standard ruler in the matter power spectrum and two-point correlation function. This paper reformulates that ruler within Darke Theory. The central claim is not that the ruler is unreal, nor that the orthodox phenomenology is false, but that the deeper load-bearing object is misidentified if it is taken to be primitively acoustic in an ontologically pre-given background. In the present framework, the observed BAO scale is the frozen remnant of a finite admissible early transport regime. Transport is not presumed to persist by default. It exists only while its burden remains supportable within the available support of the regime. The preferred scale is then the transport horizon accumulated over that admissible interval and retained in matter through a finite freeze-conversion process.

A constitutive model is developed. A transport admissibility ratio, transport activation factor, effective transport speed, running transport horizon, and finite freeze kernel are defined. A seeded perturbation equation is introduced and reduced, under spherical symmetry, to a one-dimensional radial transport equation. From this, a front law is obtained: the dominant outgoing branch is carried at the integrated transport radius $R(t)$. Finite freeze conversion then yields an effective shell with retained centre $r_{\mathrm{BAO}}^{\mathrm{DT}}$, amplitude A_{f}, and width $\sigma_{\mathrm{f,eff}}$. The resulting matter power spectrum takes the form

$$P_m(k) = P_{\mathrm{seed}}(k) \left| 1 + A_{\mathrm{f}} e^{-k^2 \sigma_{\mathrm{f,eff}}^2/2} j_0\big(k r_{\mathrm{BAO}}^{\mathrm{DT}}\big) \right|^2 ,$$

with a corresponding bump-like excess in configuration space near $r \approx r_{\mathrm{BAO}}^{\mathrm{DT}}$.

The orthodox sound-horizon picture is recovered as an effective special case when the admissible transport speed reduces to the usual sound-speed-type form and the freeze kernel is concentrated near the drag epoch. Standard cosmology therefore appears here not as something to be denied, but as a downstream effective limit of a more primitive transport-admissibility picture. The paper is constitutive and theorem-level. It does not yet fit survey data directly, nor does it derive the transport burden and support functionals from a deeper closure-dynamics law. It establishes the mathematical and structural framework that any later empirical programme would have to test.

1 Introduction

Baryon acoustic oscillations occupy a central place in modern cosmology because they furnish a preferred large-scale ruler in the clustering of matter. In the orthodox account, primordial overdensities in the tightly coupled baryon-photon plasma launch pressure-supported waves that propagate outward until baryons dynamically separate from radiation near the drag epoch. The retained scale later appears as oscillatory structure in the matter power spectrum and as a bump-like feature in the two-point correlation function, and it is used as a standard ruler for constraining cosmic expansion history [1, 4].

The present paper does not challenge that observational role. It challenges the deeper ontological reading of the ruler. Within Darke Theory, persistent structure is not primitive. It emerges only downstream of recursive differentiation, closure, admissibility, and stable carrying. That reversal of explanatory order matters. If persistent transport is itself a supportable regime rather than a default background property, then the correct foundational question is no longer merely what oscillated in the early plasma. It is why a finite transport memory survives at all, what law determines its retained scale, and what kind of structural object the observed ruler actually is.

The thesis of the paper is exact. The BAO ruler is reformulated as a frozen coherence imprint. More precisely, it is the retained shell scale left behind when a localised early disturbance launches a radially propagating transport front in a regime where transport remains admissible, and that front is later converted into persistent matter structure over a finite freeze interval. The ruler is therefore not treated here as primitively a sound horizon in an ontologically pre-given cosmological background. It is treated as the retained horizon of a finite admissible transport phase.

This shift is intended to be structurally deeper, not phenomenologically reckless. The paper does not argue that the standard BAO ruler disappears. It argues that the standard sound-horizon description should be recoverable as an effective limit of a more primitive transport-freeze picture. That is the standard of success adopted here: not denial of the orthodox observable, but reinterpretation of its constitutive meaning.

The paper proceeds as follows. Section 2 records notation and status conventions. Section 3 states the minimum Darke-Theoretic background required. Section 4 gives the primitive DT statement of BAO. Section 5 introduces the constitutive transport framework. Section 6 states the seeded transport equation. Section 7 derives the radial reduction and front law. Section 8 introduces finite freeze conversion. Section 9 compresses the resulting freeze mixture into an effective shell. Section 10 derives the oscillation law in Fourier space. Section 11 states the main interpretation theorem. Section 12 shows how the standard BAO ruler is recovered as an effective limit. Section 13 records scope and failure modes. The conclusion is brief.

2 Notation and status conventions

The paper separates four levels of statement.

(1) **Definitions.** These introduce the constitutive objects.

(2) **Assumptions.** These state the minimal regime restrictions required for the theorem chain.

(3) **Propositions, lemmas, and theorems.** These are the derived statements proved from the definitions and assumptions.

(4) **Scope limits and failure modes.** These state exactly what is not yet proved.

The principal symbols are as follows.

$\Lambda_{\mathrm{tr}}(t)$	transport admissibility ratio,
$\mathcal{A}_{\mathrm{tr}}(t)$	transport activation factor,
$v(t)$	effective transport speed,
$R(t)$	transport horizon,
$H(t)$	damping exponent,
$\mathcal{F}(t)$	finite freeze kernel,
$W(t)$	freeze weight,
$r_{\mathrm{BAO}}^{\mathrm{DT}}$	retained DT BAO ruler,
A_{f}	effective shell amplitude,
$\sigma_{\mathrm{f,eff}}$	effective shell width,
$K_{\mathrm{shell}}^{\mathrm{eff}}(r)$	effective radial shell kernel.

The present paper is constitutive and theorem-level. It is not yet a data-fitting paper. No empirical parameter values are inferred here, and no detailed comparison with survey likelihood pipelines is attempted.

3 Foundational setting

3.1 Recursive differentiation, closure, and admissibility

Darke Theory does not begin with a primitive manifold populated by already stable processes. It begins with recursive differentiation. Stable structure appears only where closure is achieved, and not every formally writable regime is supportable. Admissibility is the filter that distinguishes persistent, carried structure from merely imaginable branches. The consequence relevant here is simple: long-range transport is not to be assumed by default. It must itself be supportable.

3.2 Transport as a supportable regime

Transport is not a primitive entitlement of the regime. It is a supportable phase of the regime. In the present paper transport is treated as a regime whose continuation requires support. It carries burden. It can fail. It can switch off. A preferred scale can then arise not merely because a wave equation happens to exist, but because a transport channel exists only over a finite interval and leaves behind a retained imprint when it shuts down.

3.3 Preferred scales as freeze-retained horizons

The general expectation inside this framework is therefore the following. If a localised seed launches outward transport in a regime where transport remains admissible for a finite interval, and that transport is later frozen into a persistent matter-bearing structure, then the resulting matter field should retain a preferred shell scale equal to the accumulated transport horizon of the active interval. This is the primitive DT template into which BAO is placed.

4 Primitive DT statement of BAO

Definition 4.1 (DT statement of baryon acoustic oscillations)**.** Baryon acoustic oscillations in Darke Theory are the persistent preferred matter-separation scale generated when an early admissible radial transport regime propagates outward from localised seeds and is later frozen into matter through finite conversion.

This definition changes the explanatory order. It does not begin by positing a primitive sound process in an already neutral and already given cosmological background. It begins by identifying the observational ruler with the retained shell-scale of a finite transport phase. The ruler is thus transport-generated, limited by admissibility, and retained by freeze conversion.

5 Constitutive transport framework

Definition 5.1 (Transport admissibility ratio)**.** Let $B_{\mathrm{tr}}(t)$ be the transport burden and $S_{\mathrm{tr}}(t)$ the transport support. Define

$$\Lambda_{\mathrm{tr}}(t) := \frac{B_{\mathrm{tr}}(t)}{S_{\mathrm{tr}}(t)}. \tag{1}$$

Definition 5.2 (Transport activation factor)**.** Let $\chi(\Phi_0) \geq 0$ vanish below the relevant closure threshold and be positive above it. Let $g : \mathbb{R} \to [0, \infty)$ satisfy

$$g(z) = 0 \quad \text{for } z \leq 0, \qquad g(z) > 0 \quad \text{for } z > 0.$$

Define

$$\mathcal{A}_{\mathrm{tr}}(t) := \chi(\Phi_0(t))\, g\big(1 - \Lambda_{\mathrm{tr}}(t)\big). \tag{2}$$

Definition 5.3 (Effective transport speed)**.** Let $c_{\mathrm{coh}}(t)$ denote the coherence-transport speed in the active regime. Define

$$v(t) := \frac{c_{\mathrm{coh}}(t)\sqrt{\mathcal{A}_{\mathrm{tr}}(t)}}{a(t)}. \tag{3}$$

Definition 5.4 (Transport horizon)**.** For any $t \geq t_{\mathrm{seed}}$, define

$$R(t) := \int_{t_{\mathrm{seed}}}^{t} v(s)\, ds. \tag{4}$$

Definition 5.5 (Damping exponent)**.** Let $\Gamma(t)$ denote the damping rate. Define

$$H(t) := \frac{1}{2}\int_{t_{\mathrm{seed}}}^{t} \Gamma(s)\, ds. \tag{5}$$

Remark 5.6. The orthodox sound-horizon picture will later appear as an effective special case in which $v(t)$ reduces to the usual sound-speed-type transport law.

6 Seeded transport equation

Definition 6.1 (Seed source)**.** Let the launched seed be

$$\mathcal{S}(\mathbf{x}, t) = Q_0\, \rho_{\sigma_0}(\mathbf{x})\, \delta(t - t_{\mathrm{seed}}), \tag{6}$$

where $Q_0 > 0$ is the launched strength and ρ_{σ_0} is a normalised narrow profile satisfying

$$\int_{\mathbb{R}^3} \rho_{\sigma_0}(\mathbf{x})\, d^3x = 1. \tag{7}$$

The perturbation field $\delta\Phi(\mathbf{x}, t)$ satisfies

$$\delta\ddot{\Phi} + \Gamma(t)\delta\dot{\Phi} - v^2(t)\nabla^2\delta\Phi + \mu^2(t)\delta\Phi = Q_0\,\rho_{\sigma_0}(\mathbf{x})\,\delta(t - t_{\text{seed}}). \tag{8}$$

The mass-like term $\mu^2(t)$ is an effective restoring or confinement term.

Assumption 6.2 (Isotropic background)**.** The background is homogeneous and isotropic on the relevant scale.

Assumption 6.3 (Localised seed)**.** The seed is sharply localised near $\mathbf{x} = 0$.

Assumption 6.4 (Finite admissible transport interval)**.** There exists a finite interval on which $\mathcal{A}_{\text{tr}}(t) > 0$, and transport becomes negligible or vanishes thereafter.

Assumption 6.5 (Outgoing-front dominance)**.** The post-seed solution contains a dominant outgoing branch whose front remains identifiable through the freeze interval.

Assumption 6.6 (Narrow-shell closure)**.** During the freeze interval, the outgoing shell remains narrow enough that a Gaussian shell approximation is valid.

7 Radial reduction and front propagation

Under isotropy one may write

$$\delta\Phi(\mathbf{x}, t) = \psi(r, t), \qquad r = |\mathbf{x}|.$$

Define

$$u(r, t) := r\,\psi(r, t). \tag{9}$$

Proposition 7.1 (Radial reduction)**.** *The reduced field u satisfies*

$$u_{tt} + \Gamma(t)u_t - v^2(t)u_{rr} + \mu^2(t)u = r\,Q_0\,\rho_{\sigma_0}(r)\,\delta(t - t_{\text{seed}}). \tag{10}$$

Proof. For radial ψ,

$$\nabla^2\psi = \psi_{rr} + \frac{2}{r}\psi_r.$$

Using $u = r\psi$, one computes

$$r\left(\psi_{rr} + \frac{2}{r}\psi_r\right) = u_{rr}.$$

Substituting into (8) yields (10). □

Lemma 7.2 (Finite propagation speed)**.** *Suppose the post-seed source vanishes for $t > t_{\text{seed}}$, and suppose the initial support of u lies in $0 \le r \le r_0$. Then*

$$\operatorname{supp} u(\cdot, t) \subseteq \{\, 0 \le r \le r_0 + R(t) \,\}. \tag{11}$$

Proof. The principal part of (10) is

$$\partial_t^2 - v^2(t)\partial_r^2,$$

with characteristics satisfying

$$\frac{dr}{dt} = \pm v(t).$$

No disturbance can therefore propagate beyond the integrated characteristic distance $R(t)$. □

Proposition 7.3 (Outgoing front law)**.** *Under the localised-seed and outgoing-front assumptions, the dominant outgoing branch is concentrated near the radial front*

$$r = R(t). \tag{12}$$

Proof sketch. Apply a standard eikonal ansatz to the homogeneous part of (10). The leading-order eikonal equation is

$$\theta_t^2 - v^2(t)\theta_r^2 = 0.$$

For the outgoing branch one has

$$\theta_t + v(t)\theta_r = 0,$$

so the front characteristics obey

$$\frac{dr}{dt} = v(t).$$

Integrating from t_{seed} gives $r = R(t)$. □

Definition 7.4 (Outgoing shell profile)**.** The outgoing branch is approximated by

$$\delta\Phi(r,t) \approx Q_0\, e^{-H(t)}\, \frac{1}{4\pi R(t)^2}\, G_{\sigma(t)}(r - R(t)), \tag{13}$$

where

$$G_\sigma(x) := \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right). \tag{14}$$

8 Finite freeze conversion

Definition 8.1 (Finite freeze kernel)**.** Let $\mathcal{F}(t)$ be a nonnegative compactly supported function with support in

$$I_{\text{f}} = [t_-, t_+], \qquad t_{\text{seed}} < t_- < t_+,$$

and satisfying

$$\mathcal{F}(t) \geq 0, \qquad \int_{t_-}^{t_+} \mathcal{F}(t)\, dt = \eta_{\text{f}} > 0. \tag{15}$$

Definition 8.2 (Matter conversion law)**.** Define the matter field by

$$\delta_m(\mathbf{x}) = \delta_{\text{seed}}(\mathbf{x}) + \int_{t_-}^{t_+} \mathcal{F}(t)\, \delta\Phi(\mathbf{x}, t)\, dt. \tag{16}$$

Substituting the outgoing shell form (13) gives

$$\delta_m^{\text{shell}}(r) \approx \frac{Q_0}{4\pi} \int_{t_-}^{t_+} \mathcal{F}(t) e^{-H(t)} R(t)^{-2} G_{\sigma(t)}(r - R(t))\, dt. \tag{17}$$

Definition 8.3 (Freeze weight)**.** Define

$$W(t) := \mathcal{F}(t) e^{-H(t)} R(t)^{-2}. \tag{18}$$

Definition 8.4 (Total weighted conversion)**.**

$$\mathcal{N} := \int_{t_-}^{t_+} W(t)\, dt. \tag{19}$$

Definition 8.5 (Effective frozen radius)**.**

$$r_{\mathrm{BAO}}^{\mathrm{DT}} := \frac{1}{\mathcal{N}} \int_{t_-}^{t_+} W(t)\, R(t)\, dt. \tag{20}$$

Definition 8.6 (Effective shell amplitude)**.**

$$A_{\mathrm{f}} := Q_0\, \mathcal{N} = Q_0 \int_{t_-}^{t_+} W(t)\, dt. \tag{21}$$

Definition 8.7 (Effective shell variance)**.**

$$\sigma_{\mathrm{f,eff}}^2 := \frac{1}{\mathcal{N}} \int_{t_-}^{t_+} W(t) \Big[\sigma(t)^2 + \big(R(t) - r_{\mathrm{BAO}}^{\mathrm{DT}}\big)^2\Big] dt. \tag{22}$$

Definition 8.8 (Mean intrinsic shell variance)**.**

$$\overline{\sigma^2}_W := \frac{1}{\mathcal{N}} \int_{t_-}^{t_+} W(t) \sigma(t)^2\, dt. \tag{23}$$

Definition 8.9 (Freeze-spread variance)**.**

$$\mathrm{Var}_W(R) := \frac{1}{\mathcal{N}} \int_{t_-}^{t_+} W(t) \big(R(t) - r_{\mathrm{BAO}}^{\mathrm{DT}}\big)^2\, dt. \tag{24}$$

Lemma 8.10 (Variance decomposition)**.**

$$\sigma_{\mathrm{f,eff}}^2 = \overline{\sigma^2}_W + \mathrm{Var}_W(R). \tag{25}$$

Proof. Immediate from Definitions 8.7–8.9. □

9 Effective shell theorem

Proposition 9.1 (Effective Gaussian shell closure)**.** *Under the narrow-shell assumption, the finite-freeze mixture* (17) *is approximated by a single effective Gaussian shell*

$$\delta_m^{\mathrm{shell}}(r) \approx \frac{A_{\mathrm{f}}}{4\pi}\, G_{\sigma_{\mathrm{f,eff}}}(r - r_{\mathrm{BAO}}^{\mathrm{DT}}). \tag{26}$$

Equivalently, the effective radial shell kernel is

$$K_{\mathrm{shell}}^{\mathrm{eff}}(r) = \frac{1}{4\pi \big(r_{\mathrm{BAO}}^{\mathrm{DT}}\big)^2}\, G_{\sigma_{\mathrm{f,eff}}}(r - r_{\mathrm{BAO}}^{\mathrm{DT}}). \tag{27}$$

Proof sketch. The shell contribution is a weighted mixture of narrow Gaussians centred at $R(t)$ with widths $\sigma(t)$. Under the narrow-shell closure assumption, this mixture is well approximated by a single Gaussian with the same weighted mean and variance, namely $r_{\mathrm{BAO}}^{\mathrm{DT}}$ and $\sigma_{\mathrm{f,eff}}^2$. □

10 Fourier transform and oscillation law

For any radial kernel $K(r)$, define the Fourier transform by

$$\widetilde{K}(k) = 4\pi \int_0^\infty K(r)\, j_0(kr)\, r^2\, dr, \tag{28}$$

where

$$j_0(z) = \frac{\sin z}{z}. \tag{29}$$

Proposition 10.1 (Fourier transform of the effective shell)**.** *For the effective kernel* (27)*,*

$$\widetilde{K}^{\mathrm{eff}}_{\mathrm{shell}}(k) \approx e^{-k^2\sigma^2_{\mathrm{f,eff}}/2}\, j_0\!\big(kr^{\mathrm{DT}}_{\mathrm{BAO}}\big) = e^{-k^2\sigma^2_{\mathrm{f,eff}}/2}\frac{\sin\!\big(kr^{\mathrm{DT}}_{\mathrm{BAO}}\big)}{kr^{\mathrm{DT}}_{\mathrm{BAO}}}. \tag{30}$$

Proof sketch. Since the shell is narrow and centred near $r = r^{\mathrm{DT}}_{\mathrm{BAO}}$, one may replace r^2 by $\big(r^{\mathrm{DT}}_{\mathrm{BAO}}\big)^2$ across the narrow support to leading order. The remaining integral is the transform of a translated Gaussian shell, which gives the Gaussian envelope and spherical Bessel factor. This is the shell-transform point of the paper: a narrow spherical shell in configuration space transforms into a spherical Bessel modulation in Fourier space, which is why the factor j_0 appears in the ruler law. □

Definition 10.2 (Matter transfer law)**.** Define

$$\delta_m(\mathbf{k}) = \Big[1 + A_{\mathrm{f}}\widetilde{K}^{\mathrm{eff}}_{\mathrm{shell}}(k)\Big]\delta_{\mathrm{seed}}(\mathbf{k}). \tag{31}$$

Hence

$$P_m(k) = P_{\mathrm{seed}}(k)\,\Big|1 + A_{\mathrm{f}}\widetilde{K}^{\mathrm{eff}}_{\mathrm{shell}}(k)\Big|^2. \tag{32}$$

Substituting (30) gives

$$P_m(k) = P_{\mathrm{seed}}(k)\,\Big|1 + A_{\mathrm{f}}e^{-k^2\sigma^2_{\mathrm{f,eff}}/2} j_0\!\big(kr^{\mathrm{DT}}_{\mathrm{BAO}}\big)\Big|^2. \tag{33}$$

For $|A_{\mathrm{f}}| \ll 1$,

$$P_m(k) \approx P_{\mathrm{seed}}(k)\left[1 + 2A_{\mathrm{f}}e^{-k^2\sigma^2_{\mathrm{f,eff}}/2}\frac{\sin\!\big(kr^{\mathrm{DT}}_{\mathrm{BAO}}\big)}{kr^{\mathrm{DT}}_{\mathrm{BAO}}}\right]. \tag{34}$$

Theorem 10.3 (Oscillation theorem)**.** *Under the standing assumptions, the late matter power spectrum contains an oscillatory modulation with characteristic scale* $r^{\mathrm{DT}}_{\mathrm{BAO}}$*.*

Proof. This follows directly from the matter transfer law (31)–(34). □

Corollary 10.4 (Correlation-space bump)**.** *The two-point correlation function contains a bump-like excess near*

$$r \approx r^{\mathrm{DT}}_{\mathrm{BAO}}. \tag{35}$$

Proof. The oscillatory structure in Fourier space is Fourier dual to a localised excess in configuration space near the shell radius. □

11 Main interpretation theorem

Theorem 11.1 (BAO as a finite admissible transport-freeze observable)**.** *Within the present DT scaffold, the BAO ruler is the freeze-retained transport horizon*

$$r_{\mathrm{BAO}}^{\mathrm{DT}} = \frac{\int_{t_-}^{t_+} \mathcal{F}(t) e^{-H(t)} R(t)^{-2} R(t)\, dt}{\int_{t_-}^{t_+} \mathcal{F}(t) e^{-H(t)} R(t)^{-2}\, dt}, \tag{36}$$

where

$$R(t) = \int_{t_{\mathrm{seed}}}^{t} \frac{c_{\mathrm{coh}}(s)\sqrt{\mathcal{A}_{\mathrm{tr}}(s)}}{a(s)}\, ds. \tag{37}$$

Therefore the load-bearing invariant is not primitively sound in a pre-given background, but the horizon accumulated by an admissible transport regime and frozen into matter by finite conversion.

Proof. Proposition 7.3 identifies the moving front with the integrated transport horizon $R(t)$. Section 8 shows that finite freeze conversion retains a weighted average of those front positions, namely $r_{\mathrm{BAO}}^{\mathrm{DT}}$. Section 10 shows that this retained shell generates the oscillatory ruler in the matter power spectrum and the bump in the correlation function. The ruler is therefore the retained transport-freeze horizon. □

12 Orthodox-limit bridge

In the standard account, BAO arise from sound waves in the baryon-photon plasma and the relevant retained scale is the sound horizon at, or closely associated with, the baryon drag epoch [1, 4, 5].

Proposition 12.1 (Standard BAO as an effective limit)**.** *Suppose that over the relevant early interval*

$$\mathcal{A}_{\mathrm{tr}}(t) \approx 1, \qquad c_{\mathrm{coh}}(t) \approx c_s(t), \qquad \mathcal{F}(t) \text{ is concentrated near the drag epoch.} \tag{38}$$

Then

$$r_{\mathrm{BAO}}^{\mathrm{DT}} \approx \int_{t_{\mathrm{seed}}}^{t_{\mathrm{drag}}} \frac{c_s(t)}{a(t)}\, dt. \tag{39}$$

Proof. Under these assumptions, the effective transport speed $v(t)$ reduces to the usual sound-speed-type transport law, and the finite freeze average is concentrated near the drag epoch. The resulting ruler is therefore the standard sound-horizon-type ruler. □

This matters. The orthodox BAO description is not rejected here. It is recovered as an effective limit inside the deeper transport-admissibility structure.

13 Scope, failure modes, and exact theorem status

This paper proves a constitutive ruler law, not yet a survey-fit claim.

13.1 What has been established

The following has been established at constitutive and theorem level.

(1) A DT-native definition of BAO as a finite transport-freeze observable.

(2) A transport admissibility ratio and activation factor.

(3) A seeded perturbation equation for the early transport regime.

(4) A radial reduction and front law.

(5) A finite freeze conversion law.

(6) A closed effective shell with well-defined centre, amplitude, and width.

(7) An oscillation theorem in the matter power spectrum.

(8) A correlation-space bump corollary.

(9) An orthodox-limit bridge recovering the standard sound-horizon picture as a special case.

13.2 What remains open

The following remains open.

(1) A deeper derivation of $B_{\mathrm{tr}}(t)$ and $S_{\mathrm{tr}}(t)$ from a more primitive closure-dynamics law.

(2) A more explicit identification of $c_{\mathrm{coh}}(t)$, $\Gamma(t)$, and $\mu(t)$ in realistic early cosmology.

(3) Data-facing fitting of the effective ruler law to observed BAO data products.

(4) Extension beyond the first isotropic pass to anisotropic or environment-conditioned corrections.

(5) A more exact error analysis of the Gaussian shell closure.

13.3 Failure modes

The present framework is weakened or fails if any of the following hold.

(1) No finite admissible transport window exists.

(2) No identifiable outgoing front survives the active interval.

(3) Freeze conversion destroys shell structure rather than preserving it.

(4) The effective shell kernel fails to generate a stable oscillatory ruler.

(5) The orthodox limit cannot be recovered cleanly.

These are the correct failure conditions. The paper does not claim more than that.

14 Conclusion

This paper has reformulated baryon acoustic oscillations as a frozen coherence imprint within Darke Theory. The central move is simple but load-bearing: the BAO ruler is treated not as primitively acoustic in an ontologically pre-given background, but as the retained shell scale of a finite admissible transport regime frozen into matter through finite conversion. The preferred scale is therefore the retained transport-freeze horizon $r_{\mathrm{BAO}}^{\mathrm{DT}}$, not a primitive ruler assumed at the start.

A full constitutive chain has been given. A transport admissibility ratio and transport activation factor define when early transport is supportable. A seeded perturbation equation yields a radial outgoing front at the integrated transport radius $R(t)$. Finite freeze conversion produces an effective shell with definite retained centre, width, and amplitude. That shell, in turn, generates oscillatory modulation in the matter power spectrum together with a bump-like excess in the two-point correlation function. The standard BAO ruler is thus recovered as an observable consequence of the transport-freeze architecture rather than taken as primitive.

The orthodox sound-horizon account is not discarded. It is recovered as an effective special case when the admissible transport speed reduces to the usual sound-speed-type form and freeze is concentrated near the drag epoch. In that sense, the present framework does not attempt to erase standard BAO phenomenology. It attempts to place it downstream of a more primitive structural law.

What has been achieved is constitutive and theorem-level, not yet empirical closure. The paper does not derive the transport burden and support functionals from a deeper closure-dynamics law, and it does not yet fit survey data directly. But it does identify the exact object that a Darke-Theoretic BAO programme would have to test: the finite admissible transport-freeze horizon as the deeper invariant behind the observed standard ruler.

References

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